and triple bonds respectively. **The secondary suffix describes the functional group of the compound**.

Table 11.5 Suffix for IUPAC Name

Class of the Compound	Functional group	Suffix used
Alcohol	–OH	-ol
Aldehyde	–CHO	-al
Ketone	O ‖ –C–	-one
Carboxylic acid	–COOH	-oic acid

11.5.3 IUPAC rules for naming organic compounds:

- **Rule1:** Identify the longest chain of carbon atoms to get the parent name (root word).
- **Rule 2:** Number the carbon atoms of the parent chain, **beginning at the closest end of the substituent or functional group**. These are called **locant numbers**. *If both functional group and substituent are present, then the priority will be given to the functional group*.
- **Rule 3:** In case of alkenes and alkynes, locate the double bond or triple bond and use its locant number followed by a dash and a primary suffix. The carbon chain is numbered in such a way that the multiple bonds have the lowest possible locant number.
- **Rule 4:** If the compound contains functional group, locate it and use its locant number followed by a dash and a secondary suffix.
- **Rule 5:** When the primary and secondary suffixes are joined, the terminal 'e' of the primary suffix is removed.
- **Rule 6:** Identify the substituent and use a number followed by a dash and a prefix to specify its location and identity.

11.5.4 IUPAC Nomenclature of hydrocarbons – Solved examples

Let us try to name, systematically, some of the linear and substituted hydrocarbons by following IUPAC rules:

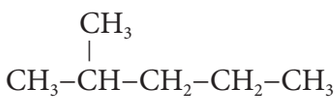
Example 1: CH_3-CH_2-CH_2-CH_2-CH_3

Step 1: It is a five- carbon chain and hence the root word is 'Pent'. (Rule 1)

Step 2: All the bonds between carbon atoms are single bonds, and thus the suffix is 'ane'.

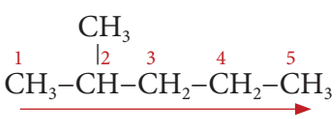
So, its name is **Pent + ane = Pentane**

Example 2:



Step 1: The longest chain contains five carbon atoms and hence the root word is 'Pent'.

Step 2: There is a substituent. So, the carbon chain is numbered from the left end, which is closest to the substituent. (Rule 2)



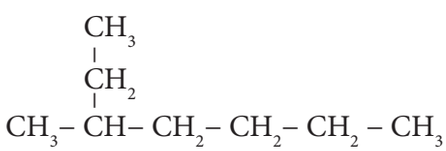
Step 3: All are single bonds between the carbon atoms and thus the suffix is 'ane'.

Step 4: The substituent is a methyl group and it is located at second carbon atom. So, its locant number is 2. Thus the prefix is '2-Methyl'. (Rule 6).

The name of the compound is

2-Methyl + pent +ane = 2-Methylpentane

Example 3:



Step 1: The longest chain contains seven carbon atoms and hence the root word is 'Hept'.

Step 2: There is a substituent. So, the

10th_Science_Unit-11.indd 161 06-12-2021 18:38:52